

Teorema de Bayes

Teorema 1 (de Bayes). Sea $(X, \Sigma, \mathbb{P})$ un espacio de probabilidad y $A, B \in \Sigma$

$$\implies \mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \cdot \mathbb{P}(A)}{\mathbb{P}(B)}.$$

Demostración: Por definición de probabilidad condicionada tenemos que

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \quad \wedge \quad \mathbb{P}(B \mid A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)}.$$

Luego despejando $\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A) \cdot \mathbb{P}(A) = \mathbb{P}(A \mid B) \cdot \mathbb{P}(B)$

$$\implies \mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \cdot \mathbb{P}(A)}{\mathbb{P}(B)}.$$

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